

Rapid terrain adaptation through continual learning of central pattern generator parameters for quadrupedal robot locomotion

Alex Pitoiset¹, Tim N. Nisslbeck¹, Ajith Anil Meera¹ and Wouter M. Kouw¹

Abstract—Central pattern generators (CPGs) produce efficient rhythmic locomotion from a small set of parameters, but a parameter set tuned for one terrain is no longer optimal once the terrain changes. We show that the CPG parameters yielding stable walking on flat ground and on a 10° incline are distinct, and that keeping the flat-optimal parameters after the transition roughly doubles the probability of a fall. Resuming stable locomotion therefore requires online adaptation. We propose an event-triggered scheme in which a prediction-error monitor detects the terrain change and initiates re-tuning of the CPG parameters within the same walking bout. Using this scheme, we compare three parameter-selection strategies, grid search, Bayesian optimization, and an autoregressive active inference agent that minimizes expected free energy. On a flat-to-incline transition over 100 trials, the active inference agent matches an oracle that switches directly to the incline optimum and keeps falls at the level of the best fixed gait, whereas grid search and Bayesian optimization roughly double the fall rate by destabilizing the robot while probing candidate gaits. Selecting parameters from a learned model, rather than by physical trial, is what allows adaptation without sacrificing stability.

I. INTRODUCTION

Achieving the same level of adaptability as animals is crucial for deploying legged machines in constantly changing real world environments. To bridge the gap between engineering and nature, engineers have frequently turned to biology for ideas, replicating animal morphologies. For example, snake-like robots for constrained environments or quadruped platforms for irregular terrain [8], [3]. These morphologies inherit the mechanical advantages of their biological counterparts. However, their control systems often lack in adaptability. For example, the Boston Dynamics quadrupedal robot, Spot, mimics animal legs. However, it relies on centralized and computationally intensive controllers that require multiple sensors [3]. Although this approach performs well, it remains complex and expensive, limiting its scalability for wider deployment and access.

To address these challenges, a promising approach to improve robotic locomotion is to use Central Pattern Generators (CPGs), which are neural circuits that generate high-dimensional rhythmic signals. In vertebrates, CPGs are located in the spinal cord and produce periodic coordinated movements such as walking and breathing [5]. By automating these repetitive physical patterns, CPGs allow the brain to redirect its resources to other cognitive tasks, such as selecting safe paths or responding to unexpected stimuli. These

systems are triggered by simple inputs from higher brain centers. The brain can modulate their rhythmic output to control parameters such as locomotion speed and gait. Moreover, CPGs remain coupled to the body and environment through sensory feedback [2]. This decentralized architecture reduces the delays for commands and the required bandwidth while simplifying high-level control [6]. These features make CPG-based controllers an attractive solution as they are less energy intensive, simpler, and more affordable.

A. Related Work

Inspired by biological principles, various CPG models have been implemented in robotics. Among them, coupled nonlinear oscillators have emerged as biologically plausible and computationally efficient for generating rhythmic movement [1], [6].

The power of coupled nonlinear oscillators lies in their ability to inhibit each other. This coupling behavior enables synchronization between joints, producing coordinated gaits and smooth transitions between them [1], [11], [9]. Such mechanisms mimic biological inter-neuronal inhibition and allow the generation of out-of-phase rhythmic signals. They also exhibit a limit-cycle behavior, producing periodic and self-sustaining oscillations that quickly recover from transient perturbations [11]. In addition, their compact dynamics, governed by a small set of parameters (e.g. frequency and amplitude), makes them appropriate for embedded systems with limited resources. Despite their promise, most existing implementations target high-end robotic platforms. Their applicability to compact, affordable systems remains underexplored, which underpins the present study.

A complementary line of work treats gait tuning as black-box optimization, adjusting the controller parameters from measured performance rather than from a model of the dynamics. Bayesian optimization is a common choice when each evaluation is expensive, as a physical or simulated walking trial is. It fits a probabilistic surrogate, typically a Gaussian process, to the parameter-score pairs observed so far, and selects the next parameters by maximizing an acquisition function such as the upper confidence bound, which favors settings that are either predicted to score highly or are still uncertain. Run *online*, the procedure scores the parameters used over the most recent time window, augments the surrogate with that observation, and proposes the parameters for the next window, so the gait is refined during walking rather than between episodes. Because unconstrained proposals can jump to destabilizing gaits, the search is confined to a trust region around the current best

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¹Department of Electrical Engineering, Eindhoven University of Technology, Netherlands w.m.kouw@tue.nl

parameters [4]. We adopt this online Bayesian optimization as one of the adaptation baselines in Section IV.

In the following, we contribute to the field with

- the design of an intelligent agent that continuously updates parameters of a central pattern generator, and
- by benchmarking this agent against offline grid search and online Bayesian optimization.

II. PROBLEM STATEMENT

We first introduce the central pattern generator that produces coordinated leg motion in a quadrupedal robot. We then argue that the parameters yielding stable locomotion depend on the terrain, so that a parameter set tuned for one terrain degrades once the terrain changes. This motivates online adaptation of the parameters as a requirement for resuming stable locomotion.

A. Central Pattern Generator

The CPGs are modeled as a network of four mutually coupled Hopf oscillators, each driving one limb: left front (LF), right front (RF), left hind (LH), and right hind (RH). Coordination between limbs arises from inhibitory coupling through a gait-specific coupling matrix $K = [k_{ij}]$. Each oscillator evolves as:

$$\dot{x}_i = \alpha(u - r_i^2)x_i - \omega_i y_i, \quad (1)$$

$$\dot{y}_i = \beta(u - r_i^2)y_i + \omega_i x_i + \gamma \sum_{j=1}^4 k_{ij} y_j, \quad (2)$$

$$\omega_i = \frac{\omega_{\text{stance}}}{1 + e^{-b y_i}} + \frac{\omega_{\text{swing}}}{1 + e^{b y_i}}, \quad (3)$$

where $r_i = \sqrt{x_i^2 + y_i^2}$, u sets the squared limit-cycle amplitude, $\alpha, \beta > 0$ are convergence rates, γ is the coupling gain, and b controls the sharpness of the frequency transition between stance and swing phases.

The CPG outputs (x_i, y_i) are mapped to hip and knee joint angles:

$$\phi_{\text{hip}} = \phi_{\text{hip}}^0 + A_{\text{hip}} x_i, \quad (4)$$

$$\phi_{\text{knee}} = \phi_{\text{knee}}^0 - A_{\text{knee}} \max(0, y_i), \quad (5)$$

where x_i drives the forward-backward hip swing and y_i controls knee flexion, the fixed offsets ϕ_{hip}^0 and ϕ_{knee}^0 set the neutral pose, the gains A_{hip} and A_{knee} scale the swing and flexion amplitudes, and negative y_i is suppressed to prevent hyperextension.

The behavior of the CPG is governed by a small set of parameters, collected in the vector

$$\theta = [\gamma, \omega_{\text{swing}}, \omega_{\text{stance}}, F_{\text{fast}}, K_{\text{stop}}, A_{\text{hip}}, A_{\text{knee}}, b], \quad (6)$$

comprising the coupling gain, the phase-dependent stance and swing frequencies, a fast-contact gain and stop gain from the sensory feedback, the hip and knee amplitudes, and the sigmoid sharpness b of Eq. 3. These parameters set the gait shape, and are held fixed during locomotion.

B. Terrain-dependent parameters

For a fixed terrain, the parameter vector θ can be tuned to produce stable locomotion. During a bout the robot reports its forward velocity v_x and the roll and pitch of its trunk, ψ_r and ψ_p , which measure the deviation from an upright attitude. Let e denote the terrain. We score a parameter set over an evaluation window by

$$V(\theta, e) = \min\left(\frac{\bar{v}_x}{v^*}, 1\right) - \frac{1}{\psi_0} \left(\text{RMS}[\psi_r] + \text{RMS}[\psi_p] \right), \quad (7)$$

and set $V = -c_f$ whenever the robot falls. Here $\bar{v}_x$ is the mean forward velocity, v^* the target speed, and ψ_0 normalizes the attitude penalty; the root-mean-square is taken over the window on low-pass-filtered angles, so that the gait's rhythmic rocking is not penalized, and $\tilde{\psi}_p$ is the pitch measured relative to the terrain incline. The progress term saturates at v^* , so moving slower up an incline or faster down it carries no penalty as long as the robot keeps advancing toward its goal, while the attitude term dominates the score, encoding that stable locomotion is the primary requirement. The best gait for terrain e is then the terrain-conditioned optimum

$$\theta^*(e) = \arg \max_{\theta} V(\theta, e). \quad (8)$$

The difficulty is that $\theta^*(e)$ is not constant across terrains. A gait that keeps the trunk level on flat ground pitches the body, shifts foot-contact timing, and alters the coupling needed for coordination once the same legs meet an incline, so the location of the optimum moves. We consider the two terrains of Fig. 1: flat ground and a 10° incline. Optimizing Eq. 7 on each terrain with Bayesian optimization yields the parameter vectors shown in the lower panel, and they are visibly different. The incline optimum uses markedly stronger inter-limb coupling, with the coupling gain γ raised to the top of its range against the bottom of its range on flat ground, together with a much softer stance-to-swing frequency transition, expressed by a small sharpness b ; the stance and swing frequencies and the contact-feedback gains stay comparable. Each set is more stable on the terrain it was tuned for, and retaining the flat-optimal parameters after the ground turns into an incline roughly doubles the probability of a fall relative to adopting the incline-optimal parameters (Section IV).

Because θ is held fixed, a change in terrain leaves the controller operating at a now-suboptimal point $\theta^*(e_{\text{old}}) \neq \theta^*(e_{\text{new}})$. The resulting attitude deviations compound over time: as the trunk tilts away from balance, recovery demands larger corrections, and locomotion can fail before the robot has traversed the new terrain. Fig. 2 shows a representative bout. While the terrain is flat the trunk stays within a couple of degrees of vertical, but when the robot meets the incline with the flat-optimal parameters unchanged, its tilt grows over a few seconds until it topples and forward progress stops. Switching to the incline-optimal parameters at the transition instead holds the tilt near the modest baseline set by the incline, and the robot keeps climbing. To resume stable

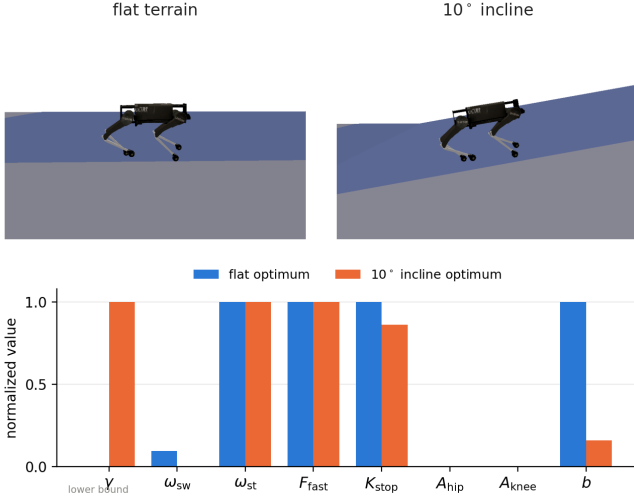


Fig. 1. The optimal CPG parameters depend on the terrain. *Top*: the Laikago walking on flat ground and on a 10° incline. *Bottom*: the parameter vectors that maximize the stability criterion (Eq. 7) on each terrain, normalized to the shared bounds. The optima differ, most visibly in the coupling gain γ and the frequency-transition sharpness b .

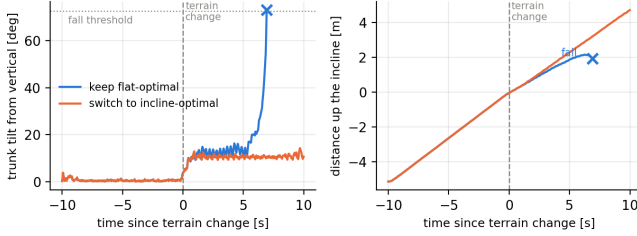


Fig. 2. Keeping the flat-optimal parameters after a terrain change destabilizes the robot. *Left*: the trunk tilt from vertical stays small on flat ground; after the transition (dashed line) it settles near the incline baseline when the parameters are switched to the incline optimum, but grows past the fall threshold when they are kept. *Right*: the robot climbs steadily after switching, whereas keeping the flat-optimal parameters ends in a fall and halted progress. Curves are one representative seed.

locomotion after a terrain change, the CPG parameters must therefore be adapted online, which requires both detecting that the terrain has changed and re-tuning θ toward $\theta^*(e_{new})$ quickly enough to preserve balance.

III. METHODS

A. Autoregressive Active Inference Agent

The agent is based on a multivariate autoregressive model with exogenous inputs (MARX) [7]. It maintains a belief over dynamics parameters updated from observations, and selects controls by minimising an Expected Free Energy (EFE) objective over a planning horizon.

1) *Model*: The likelihood is a Gaussian autoregression with coefficient matrix $A \in \mathbb{R}^{D_x \times D_y}$ and noise precision $W \in \mathbb{R}_+^{D_y \times D_y}$. Denoting $\Theta = (A, W)$:

$$p(y_k | \Theta, u_k, \bar{u}_k, \bar{y}_k) = \mathcal{N}(y_k | A^T x_k, W^{-1}), \quad (9)$$

where $\bar{u}_k$ and $\bar{y}_k$ are running memories of the M_u previous inputs and M_y previous outputs, and the regressor concatenates

the current control with the memories:

$$x_k = \begin{bmatrix} u_k \\ \bar{u}_k \\ \bar{y}_k \end{bmatrix}, \quad D_x = D_u(M_u + 1) + D_y M_y. \quad (10)$$

The prior over parameters is a matrix-normal-Wishart (MNW) distribution [10, ID: P323]:

$$p(\Theta) = \mathcal{M}\mathcal{N}(A | M_0, \Lambda_0^{-1}, W^{-1}) \mathcal{W}(W | \Omega_0^{-1}, v_0). \quad (11)$$

Controls are given an independent Gaussian prior for regularisation, with precision Υ :

$$p(u_k) = \mathcal{N}(u_k | 0, \Upsilon^{-1}). \quad (12)$$

The target observation is encoded as a Gaussian goal prior:

$$p(y_t | y_*) = \mathcal{N}(y_t | m_*, S_*). \quad (13)$$

2) *Parameter Estimation*: Because the MNW distribution is conjugate to the Gaussian likelihood, Bayesian filtering yields a closed-form MNW posterior $p(\Theta | \mathcal{D}_k) = \mathcal{M}\mathcal{N}\mathcal{W}(A, W | M_k, \Lambda_k^{-1}, \Omega_k^{-1}, v_k)$ at each step k , where $\mathcal{D}_k = \{y_i, u_i\}_{i=1}^k$. The parameters are updated by rank-one corrections:

$$v_k = v_{k-1} + 1, \quad (14)$$

$$\Lambda_k = \Lambda_{k-1} + x_k x_k^T, \quad (15)$$

$$M_k = (\Lambda_{k-1} + x_k x_k^T)^{-1} (\Lambda_{k-1} M_{k-1} + x_k y_k^T), \quad (16)$$

$$\Omega_k = \Omega_{k-1} + y_k y_k^T + M_{k-1}^T \Lambda_{k-1} M_{k-1} - (\Lambda_{k-1} M_{k-1} + x_k y_k^T)^T (\Lambda_{k-1} + x_k x_k^T)^{-1} (\Lambda_{k-1} M_{k-1} + x_k y_k^T).$$

Marginalising the likelihood over the MNW prior yields a multivariate Student-t posterior predictive:

$$p(y_k | u_k, \mathcal{D}_{k-1}) = \mathcal{T}_{\eta_k}(y_k | \mu_k(u_k), \Sigma_k(u_k)), \quad (17)$$

with $\eta_k = v_{k-1} - D_y + 1$ and

$$\mu_k(u_k) = M_{k-1}^T x_k, \quad (18)$$

$$\Sigma_k(u_k) = \frac{\Omega_{k-1}}{v_{k-1} - D_y + 1} \left(1 + x_k^T \Lambda_{k-1}^{-1} x_k \right). \quad (19)$$

Both the mean μ_k and scale Σ_k are smooth functions of u_k .

3) *Planning*: To select controls over a horizon $\tau = (k+1, \dots, k+T)$, the agent unrolls the autoregressive model forward. At each step $t \in \tau$, the regressor is formed as $x_t = [u_t, \bar{u}_t, \hat{y}_{t-1}]^T$, where future outputs not yet observed are substituted with MAP estimates of earlier predictives, $\hat{y}_{t-1} = [\mu_{t-1}(u_{t-1}), \mu_{t-2}(u_{t-2}), \dots]$. This yields a sequence of Student-t predictives $p(y_t | u_t, \mathcal{D}_k)$ with means $\mu_t(u_t)$ and scales $\Sigma_t(u_t)$.

The planning objective is the Expected Free Energy (EFE), defined as the sum of KL divergences between each posterior predictive and the goal prior:

$$\mathcal{J}(u_\tau) = \sum_{t \in \tau} D_{\text{KL}}[p(y_t | u_t, \mathcal{D}_k) \parallel p(y_t | y_*)]. \quad (20)$$

Expanding this KL divergence yields goal-seeking terms (functions of μ_t , penalising deviation from m_*) and information-seeking terms (functions of Σ_t , penalising output

uncertainty). Together with the control prior, these produce the MAP action selection problem over the planning horizon:

$$\hat{u}_\tau = \arg \min_{u_\tau \in \mathbb{U}^T} \left[-\frac{1}{2} \ln |\Sigma_t(u_t)| + \frac{1}{2} \text{tr} \left[S_t^{-1} \frac{\eta_t}{\eta_t - 2} \Sigma_t(u_t) \right] + \frac{1}{2} (\mu_t(u_t) - m_*)^\top S_*^{-1} (\mu_t(u_t) - m_*) + u_t^\top \Upsilon u_t \right], \quad (21)$$

where $\mathbb{U}^T = \mathbb{U} \times \dots \times \mathbb{U}$ is the space of affordable controls over the horizon.

IV. EXPERIMENTS

We test whether *event-triggered* online adaptation of the CPG parameters can restore stable locomotion after a terrain change, and we compare three parameter-selection strategies as the adaptation mechanism: a *grid search* that samples the parameter space by Latin-hypercube design, online *Bayesian optimization* (BO) with a Gaussian-process surrogate and a UCB acquisition function, and the proposed *MARX-EFE* agent of Section III. Grid search and BO are black-box optimizers that can only assess a candidate gait by walking on it, whereas MARX-EFE selects parameters from its continually updated model without such trials. The comparison targets the transient after the terrain change: the hypothesis is that the method which finds better parameters faster keeps the robot upright, because a growing attitude deviation is progressively harder to correct.

A. Setup

Experiments use the Laikago quadruped in PyBullet at a control rate of 100 Hz ($\Delta t = 0.01$ s). The robot walks along a corridor that is flat for the first stretch and then rises at a 10° incline. It starts with the flat-optimal parameters $\theta^*(\text{flat})$ and, after 10 s of flat walking, meets the incline; the incline is placed at the robot's forward position at that instant, measured per seed by a calibration run, so that every run enters the slope in the same phase of its gait. The reference optima $\theta^*(\text{flat})$ and $\theta^*(10^\circ)$ are obtained beforehand by maximizing the criterion of Eq. 7 on each terrain with Bayesian optimization, using $v^* = 0.5$ m/s, $\psi_0 = 10^\circ$, and a fall penalty $c_f = 2$.

We run $N = 100$ seeds. Across-seed variability comes only from a small perturbation of the initial joint angles ($\sigma = 0.002$ rad), which the contact dynamics amplify into distinct trajectories; the start pose and the terrain geometry are otherwise identical across seeds, which isolates the response to the terrain change from the choice of starting point.

B. Detecting the terrain change

Adaptation is triggered by a prediction-error monitor. A MARX model (Eq. 9) is updated every control step from the observation $y_k = (v_x, v_y, \psi_p, \psi_r)$ and predicts the observation 0.5 s ahead under the current parameters. On flat ground this rollout error is small; when the robot meets the incline the prediction breaks down and the error rises. We compare an exponential moving average of the error against a baseline collected during flat walking, and fire the trigger when it exceeds baseline mean + $K \cdot$ baseline s.d. One-step innovations

TABLE I
EVENT-TRIGGERED ADAPTATION ON THE FLAT-TO- 10° TRANSITION
($N = 100$ SEEDS). LOWER IS BETTER EXCEPT UPRIGHT TIME AND $\bar{V}$.
BEST ONLINE METHOD IN BOLD.

Metric	reference		online method		
	no adapt	oracle	grid	BO	MARX-EFE
Falls, 20 s [%]	40	47	83	90	39
Falls, 10 s [%]	19	8	37	37	10
Upright time [s]	16.7	17.3	12.3	12.2	17.8
Mean $\bar{V}$	-0.36	-0.30	-0.54	-0.54	-0.33

at 100 Hz are dominated by persistence and barely register the terrain change, whereas the 0.5 s rollout compounds the model mismatch into a clear spike. Calibrating on held-out seeds, $K = 16$ is the smallest threshold with no false positives on flat-only runs and no missed detections on flat-to-incline runs; it fires on average 0.33 s after the trunk crosses onto the incline.

C. Adaptation protocol

When the trigger fires, the selected method begins proposing CPG parameters, one candidate per 1.5 s window, ramped in over 0.3 s, and each window is scored by Eq. 7. All three optimizers search a trust region of ± 0.25 of the parameter range around the incumbent and share a common safety layer: the best-scoring parameters seen so far are restored for one window whenever a candidate scores well below them. Without both the trust region and this safeguard, every method, MARX-EFE included, destabilizes the robot within seconds, so these two elements are applied uniformly across methods. MARX-EFE uses a forgetting factor of 0.995 so that its model tracks the change, and its control prior is centered on the current parameters.

We compare five arms, all sharing an identical pre-trigger trajectory per seed so that the event fires at the same instant in every arm:

- *no adaptation*: keep $\theta^*(\text{flat})$, a lower reference;
- *oracle*: switch directly to $\theta^*(10^\circ)$, an upper reference that presumes the new optimum is known;
- *grid*, *BO*, and *MARX-EFE*: the three online methods.

We report the fall rate over the 20 s after the trigger, the fall rate over the first 10 s (the transition phase, where the parameter mismatch itself is the hazard), the mean time the robot stays upright, and the mean windowed criterion $\bar{V}$.

D. Results

Table I and Fig. 3 summarize the comparison. First, the shift in the optimum is real and it matters at the transition: within the first 10 s the oracle falls in 8% of seeds against 19% for no adaptation (McNemar $p = 0.03$), so switching to terrain-matched parameters roughly halves the early falls. This is the operational counterpart of the terrain-conditioned optimum in Section II-B.

Second, MARX-EFE matches this protection without knowing the new optimum. Its early fall rate of 10% is statistically indistinguishable from the oracle and from no

Event-triggered CPG adaptation on flat→10° slope (100 seeds, adaptation window 20 s)

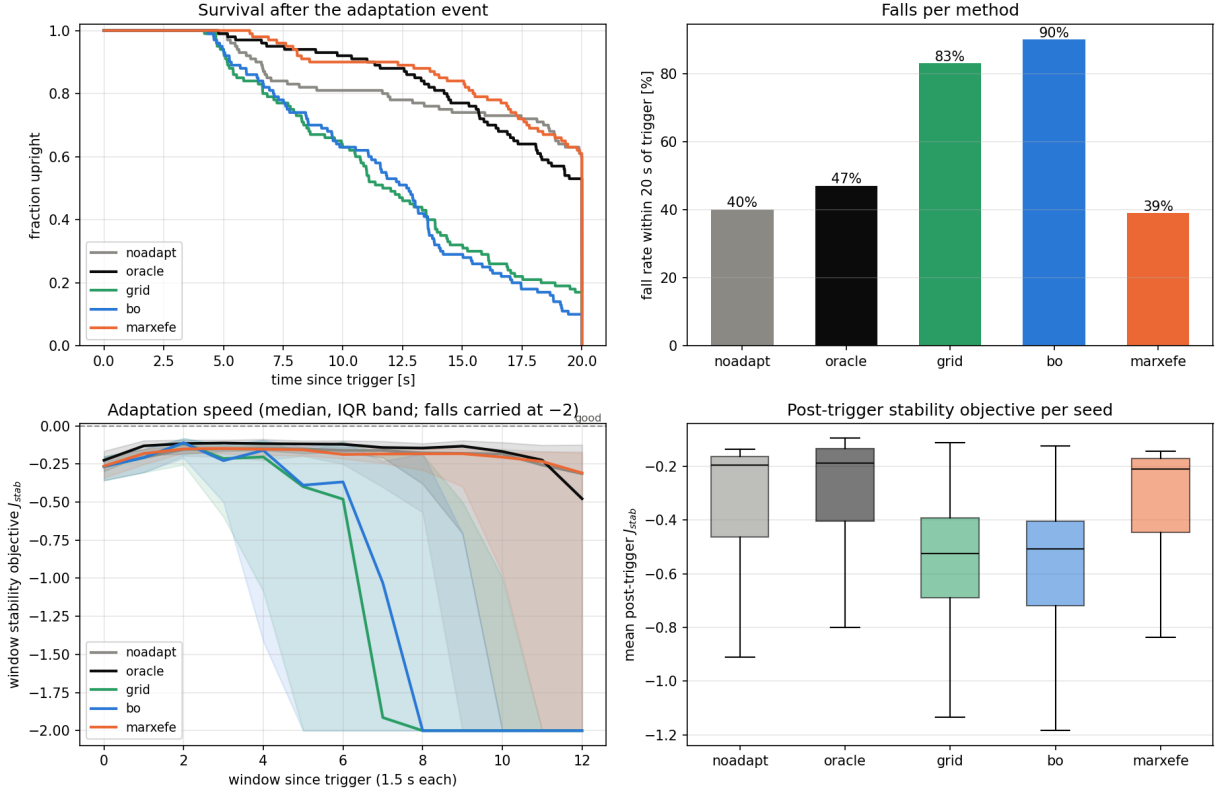


Fig. 3. Event-triggered adaptation on the flat-to-10° transition (100 seeds). *Top left*: fraction of seeds still upright versus time since the trigger. *Top right*: fall rate within 20 s. *Bottom left*: median windowed criterion V after the trigger (shaded band: interquartile range; falls carried at -2). *Bottom right*: distribution of the per-seed mean V . MARX-EFE (orange) tracks the oracle (black), while grid (green) and BO (blue) fall far more often.

adaptation, and far below grid and BO at 37% each ($p < 10^{-5}$). Over the full 20 s, MARX-EFE (39%) again matches no adaptation (40%) and stays well below grid (83%) and BO (90%); the difference against both black-box optimizers is large and significant (McNemar $p \leq 4 \times 10^{-10}$, upright-time Wilcoxon $p \approx 10^{-12}$).

Third, trial-based search is destructive when run online. Grid and BO must spend whole windows walking on speculative candidates, and even with the trust region and the safeguard this roughly doubles the fall rate relative to not adapting at all ($p < 10^{-11}$). They pay a stability price during exploration that MARX-EFE avoids by selecting parameters from its model rather than by physical trial. The bottom-left panel of Fig. 3 shows the mechanism: the windowed criterion of grid and BO collapses toward the fall value within a few windows, whereas MARX-EFE and the oracle hold a near-constant score.

Finally, over the full horizon the two references converge (no adaptation 40% versus oracle 47%, not significant): sustained 10° climbing carries a steady fall hazard for any fixed parameter set, and survivors still exhibit 12–14° peak roll. A single correct switch therefore buys survival through the transition rather than immunity thereafter, which makes the early-window contrast the more informative endpoint.

V. DISCUSSION

Our results separate two aspects of online adaptation that are easily conflated: deciding *when* to change the gait and deciding *what* to change it to. The prediction-error trigger resolves the first cheaply and reliably, firing within a third of a second of the terrain change and never on flat ground. The second is where the methods diverge. Grid search and Bayesian optimization can only judge a candidate gait by walking on it, and on an incline a poor candidate does not merely score badly, it tips the robot over; even confined to a trust region and protected by a revert-to-best safeguard, this exploration roughly doubles the fall rate relative to not adapting at all. The active inference agent instead selects parameters from a model that it updates every control step, so it avoids destabilizing physical trials and matches an oracle handed the incline optimum in advance. The advantage is therefore not faster search in the usual sense, but the ability to improve the gait *without* risking it.

Several limitations qualify these conclusions. We study a single, clean transition from flat ground to a 10° incline in simulation, with the reference optima computed offline; broader terrains, friction changes, and hardware experiments remain to be done. The open-loop CPG also carries an intrinsic hazard on a sustained slope, so that even the oracle

falls in nearly half of the long runs and the benefit of a correct switch is concentrated in the seconds around the transition. Finally, the comparison equalizes caution across methods through a shared trust region and safeguard; how far these can be relaxed for a model-based agent, which does not need physical trials to avoid harmful gaits, is an open question. Learning the terrain-conditioned optimum online, rather than assuming it is known, and coupling the agent to closed-loop attitude feedback are natural next steps.

VI. CONCLUSIONS

We showed that the CPG parameters producing stable quadruped locomotion shift with the terrain, so that a gait tuned for flat ground becomes unstable on an incline, and that resuming stable walking requires online adaptation. We proposed an event-triggered scheme in which a prediction-error monitor detects the terrain change and an adaptation method re-tunes the CPG within the same walking bout, and we compared grid search, Bayesian optimization, and an autoregressive active inference agent as that method. On a flat-to-incline transition over 100 trials, the active inference agent matched an oracle with prior knowledge of the incline optimum and roughly halved the fall rate relative to the black-box optimizers, which destabilize the robot while probing candidate gaits. Selecting parameters from a learned model, rather than by physical trial, is what makes online adaptation possible without sacrificing stability.

APPENDIX

ACKNOWLEDGMENT

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